

Senior engineer with 10+ years building award-winning systems across fintech, healthcare, and national security.

Founder & AI Engineer ▪ Spirit & Hammer

October 2025 - Present

Technologies: Next.js, React, TypeScript, Python, FastAPI, Vercel AI SDK, OpenRouter, Claude, Three.js, Cloudflare, Swift, Playwright, LLM

- Architected full-stack features and end-to-end AI agent harness orchestration, MCP server integration, and skills for security startup [Intruex](#) to win the company's first contract.
- Shipped [Sun Signal](#), a circadian-health platform delivering real-time daily timing guidance for thousands of visitors for any US ZIP, built on a 50K-LOC FastAPI engine, AI chat, and dynamic Vercel AI UI components.
- Architected an AI video pipeline & YouTube player system protecting user agency and attention and reducing YouTube production time by 50%.
- Delivered a personal brand for [Hans Morningstar](#) with blog, newsletter, and lead-capture and attribution deployed within a sandboxed environment capturing leads and enabling the client to develop his own website.
- Engineered agentic-first offline-first apps (<https://www.mannan.is/garden#products>) on Vercel & Cloudflare with encryption, distributed databases, offline functionality, emails, newsletters, and custom HMAC magic-link auth.

Principal Software Engineer ▪ Capital One

Jan 2021 to Dec 2021, Aug 2024 to Oct 2025

Technologies: Java, JavaScript, TypeScript, SolidJS, React, Angular, Spring Boot, AWS, Docker, Git, REST APIs, JUnit, AI Tools, Copilot

- Generated millions in revenue by delivering Auto Navigator Trade-In in two months with a lean team, aligning cross-functional partners and architecting scalable document upload.
- Built full-stack features (Trade-In, Auto Loan Rates, Search) serving millions of users with React, SolidJS, Angular, and Spring Boot microservices on AWS.
- Cut code-review cycles 40% and eliminated 100K API failures by leading cybersecurity remediation as Duty Champion with automated security testing.
- Enabled zero-defect deployments for 3M+ daily users modernizing legacy Angular codebases with automated testing and releases.
- Won a company-wide hackathon for UX design and frontend implementation.

Senior Software Engineer ▪ Publicis Sapient

April 2022 to May 2023

Technologies: Angular, TypeScript, Java, Spring Boot, AWS, ArcGIS, Kubernetes, REST APIs, JUnit

- Built interactive healthcare facility mapping using ArcGIS and Angular, enabling healthcare officials to map out providers across regions - increasing healthcare access by 20% across underserved regions.
- Led code review standardization and mentorship programs, reducing defect rates by 33% and establishing quality benchmarks adopted company-wide.
- Designed and implemented Angular Material UI component library for consistent cross-team design patterns, accelerating feature delivery timelines.

Senior Software Engineer ▪ Maxar Technologies

Dec 2018 to Apr 2022

Technologies: Angular, TypeScript, Java, Spring Boot, AWS, Kubernetes, REST APIs, JUnit

- Supported national decision-makers by engineering distributed Angular apps for geospatial mapping and probability simulation.
- Enabled rapid analysis of high-volume data by building real-time visualization with parallel processing in modern JS frameworks.
- Drove adoption of complex engineering solutions by partnering with technical and business stakeholders to validate requirements and deliver technical demos.

Software Engineer ▪ NRECA / America's Electric Cooperatives

Jan 2015 to Dec 2016

Technologies: Python, Flask, JavaScript, AWS, Kubernetes, REST APIs, JUnit

- Built Python/Flask Open Modeling Framework models and security-awareness simulations for electric co-op planning.

Certifications: Claude Certified Architect (July 2026), Johns Hopkins Agentic AI, AWS Certified Solutions Architect**AI & ML:** Agentic AI, RAG Pipelines, ML, LLM, Claude, Claude Code, OpenRouter, AI SDK**Languages:** Java, Kotlin, TypeScript, JavaScript, Python, Node.js**Frontend:** React, Angular, SolidJS, RxJS & NgRx, geospatial mapping (ArcGIS, Cesium.js, Leaflet.js, Google Maps API)**Backend & Data:** Spring Boot, Flask, REST, Microservices, PostgreSQL, MongoDB, NoSQL, distributed systems, parallel computing**Cloud, DevOps & Testing** : AWS (Certified Solutions Architect), Docker, Kubernetes, CI/CD, Git, JUnit/Jest/Selenium, TDD/BDD

Bachelor of Science, Electrical Engineering